

每一次 SEND 、 WAIT 、 SLEEP 都是能源決策

VERIFIED



節能是一條可檢驗的因果鏈

VERIFIED

六段箭頭流程圖。

observation → policy → action → state / packet → service → J 。

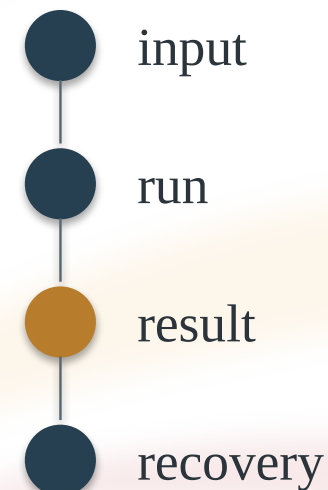
LoRaEnergySim 為什麼直接服務智慧節能與 IoT

VERIFIED

copyable operation

低功耗節點、封包與重試、睡眠／處理／TX／RX、可觀察結果。

state / receipt



LEO 只提供會改變的服務窗口

VERIFIED; DONOR-
REWRITE : BeamShift donor
slides 2, 4–6 的 changing
service window , 改寫成
endpoint-first IoT example 。

一條 NTPU contact-window timeline 。

機會出現 → policy
選擇 → service /
energy 取捨

hero timeline

LEO = example 。
VERIFIED; DONOR-
REWRITE : BeamShift
donor slides 2, 4–6 的
changing service
window , 改寫成
endpoint-first IoT
example 。

verdict

recovery / transfer

兩層 evidence 不能互相冒充

VERIFIED

endpoint replay 與 C-120 system replay 的分層圖。

**endpoint layer: queue / packet /
state / endpoint J**

split compare

**system layer: 既有 C-120
authority。**

VERIFIED

從預測走到可反駁的結果

VERIFIED

predict → edit → run → import → replay → withheld 圓環。

先寫預測，再執行

withheld case 不准 retune 。

一個小改動，必須能追到一個機制

VERIFIED

student_policy.py 的 marked block 卡片。

只改 marked block

single annotated card

policy hash → result
→ replay 。

VERIFIED

verdict

recovery / transfer

今天的教學路徑

VERIFIED

十個節點的路徑圖。

why → setup → endpoint model → anchor → A → B → recovery → C → evidence → transfer 。

EXPECT | consequential diff across state / packet / service / J

STOP | identity / unit / service 不一致就停止

RECOVER | 若 claim 被擴大，停在分層圖，回到 endpoint/system boundary

開場檢查點：先確認 claim 邊界

VERIFIED

四層 claim ladder 。

**NOT LIVE / NOT
MEASURED / NOT
CANONICAL-
PARITY-
VERIFIED 。**
VERIFIED

**course-packaged
simulated endpoint-
energy lab**

claim shield

verdict

recovery / transfer

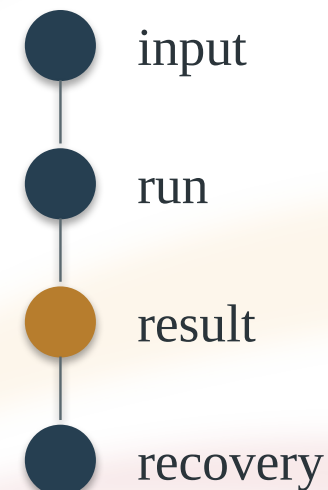
套件根目錄是一條 provenance 邊界

VERIFIED (server ZIP
inventory : single root 、 65
files)

copyable operation

README 、 setup.sh / setup.cmd 、 course.sh /
course.cmd 、 student_policy.py 、 schemas/ 。

state / receipt



EXPECT | consequential diff across state / packet / service / J

STOP | identity / unit / service 不一致就停止

RECOVER | 若 shell 、 版本 、 lock 或 receipt 不符，保留錯誤並回到 setup gate

下載指定 archive ，先核對 identity

VERIFIED (server
bytes 、 checksum 內容與
listener) ； EXTERNAL GET
REHEARSAL PENDING

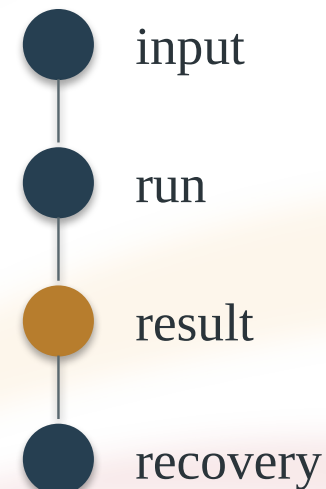
copyable operation

<http://120.126.151.102:4192/c120-lora-energy-lab-v1.zip>

<http://120.126.151.102:4192/c120-lora-energy-lab-v1.zip.sha256>

SHA-
256:ae887f751b93dbcbdb2289b0593fc2cceb01da05e2e7034db0c24f
692bf46e 。

state / receipt



Windows 原生先確認 shell 與 launcher

IMPLEMENTED / NOT
VERIFIED

CMD 與 PowerShell 的提示字元對照卡。

C:\...> → py -3.11

split shell

PS C:\...> → .\setup.cmd 。

IMPLEMENTED NOT VERIFIED

EXPECT | consequential diff across state / packet / service / J

STOP | IMPLEMENTED / NOT VERIFIED ；保留錯誤

RECOVER | 若 shell、版本、lock 或 receipt 不符，保留錯誤並回到 setup gate

沒有 Python 3.11 時的 Windows 安裝路徑

IMPLEMENTED / NOT
VERIFIED (official Windows
download path ; clean install
pending)

Python 3.11 installer → launcher → version check ◦

**[https://
www.python.org/
downloads/windows/](https://www.python.org/downloads/windows/)**

Python 3.11.x /
Windows installer (64-
bit)

安裝 Python
Launcher

重新開啟 shell → py
--list ◦

uv 是 Python 3.11 的 recovery 工具

IMPLEMENTED / NOT
VERIFIED

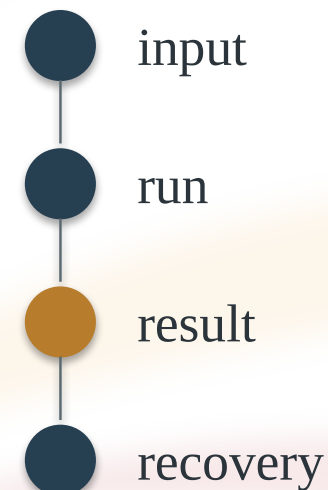
copyable operation

```
winget install --id=astral-sh.uv -e
```

```
uv python install 3.11
```

```
uv python find 3.11 ◦
```

state / receipt



EXPECT | consequential diff across state / packet / service / J

STOP | IMPLEMENTED / NOT VERIFIED ; 保留錯誤

RECOVER | 若 shell、版本、lock 或 receipt 不符，保留錯誤並回到 setup gate

Windows 用 package-local .venv

VERIFIED

system Python 與 .venv\Scripts\python.exe 的隔離圖。

system Python $\neq$ course .venv

environment split

不要共享 WSL .venv 。

VERIFIED

EXPECT | consequential diff across state / packet / service / J

STOP | identity / unit / service 不一致就停止

RECOVER | 若 shell、版本、lock 或 receipt 不符，保留錯誤並回到 setup gate

Windows 執行 setup 與 verify

IMPLEMENTED / NOT
VERIFIED

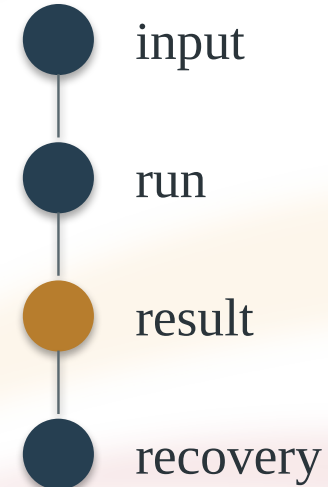
copyable operation

```
$env:PYTHON_BIN = "py -3.11"
```

```
.\setup.cmd
```

```
.\course.cmd verify
```

state / receipt



EXPECT | READY receipt

STOP | IMPLEMENTED / NOT VERIFIED ; 保留錯誤

RECOVER | 若 shell、版本、lock 或 receipt 不符，保留錯誤並回到 setup gate

WSL 先確認它真的是 Ubuntu shell

IMPLEMENTED / NOT
VERIFIED

PowerShell 安裝／清單與 Ubuntu pwd / uname -a 的雙視窗。



EXPECT | consequential diff across state / packet / service / J

STOP | IMPLEMENTED / NOT VERIFIED ; 保留錯誤

RECOVER | 若 shell、版本、lock 或 receipt 不符，保留錯誤並回到 setup gate

WSL 安裝工具與 uv

IMPLEMENTED / NOT
VERIFIED

apt → curl → uv → Python 3.11 階梯。

sudo apt update

sudo apt install -y curl unzip

curl -LsSf
<https://astral.sh/uv/install.sh> |
sh 。

EXPECT | consequential diff across state / packet / service / J

STOP | IMPLEMENTED / NOT VERIFIED ；保留錯誤

RECOVER | 若 shell、版本、lock 或 receipt 不符，保留錯誤並回到 setup gate

WSL 解壓與建立 .venv

IMPLEMENTED / NOT
VERIFIED

/mnt/c/... → ~/c120-course/ → package root ◦

```
PYTHON_BIN="$(uv python find  
3.11)"
```

.venv/bin/python ◦

bash setup.sh

IMPLEMENTED NOT VERIFIED

EXPECT | consequential diff across state / packet / service / J

STOP | IMPLEMENTED / NOT VERIFIED ; 保留錯誤

RECOVER | 若 shell、版本、lock 或 receipt 不符，保留錯誤並回到 setup gate

POSIX runner 的最小入口

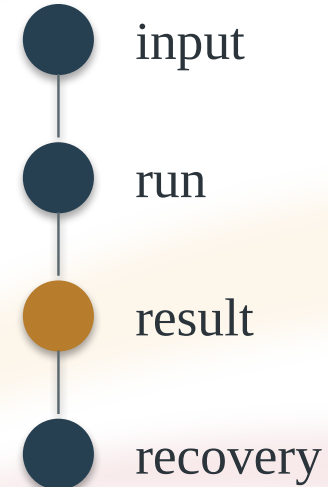
IMPLEMENTED / NOT
VERIFIED

copyable operation

```
PYTHON_BIN=python3.11 bash setup.sh
```

```
bash course.sh verify ◦
```

state / receipt



EXPECT | consequential diff across state / packet / service / J

STOP | IMPLEMENTED / NOT VERIFIED ; 保留錯誤

RECOVER | 若 shell、版本、lock 或 receipt 不符，保留錯誤並回到 setup gate

READY receipt 是下一步的入場券

EVIDENCE PLACEHOLDER

machine-readable receipt 欄位放大卡。

observation	action	mechanism	evidence
status: READY	Python 3.11.x	scenario_id	policy_api_version

setup 失敗不等於 energy result

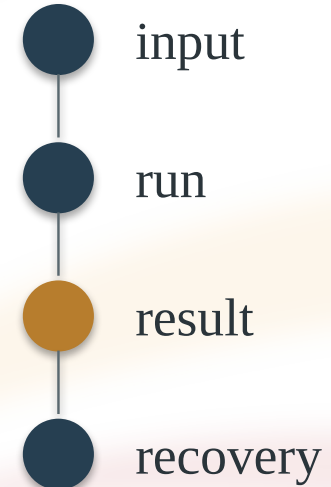
VERIFIED

copyable operation

讀錯誤 → 修最小邊界

無法完成 → same-scenario fallback ◦

state / receipt



EXPECT | consequential diff across state / packet / service / J

STOP | identity / unit / service 不一致就停止

RECOVER | 若 shell、版本、lock 或 receipt 不符，保留錯誤並回到 setup gate

可編輯檔案只有一個

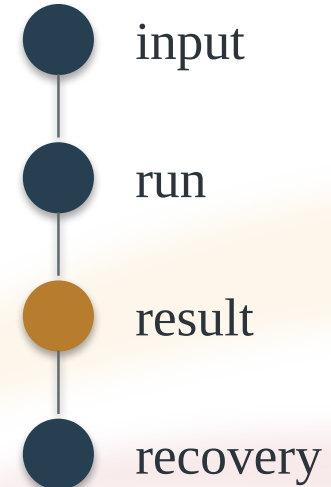
VERIFIED

copyable operation

只編輯 marked blocks

scenario / schemas / runner / generated JSON = read-only ◦

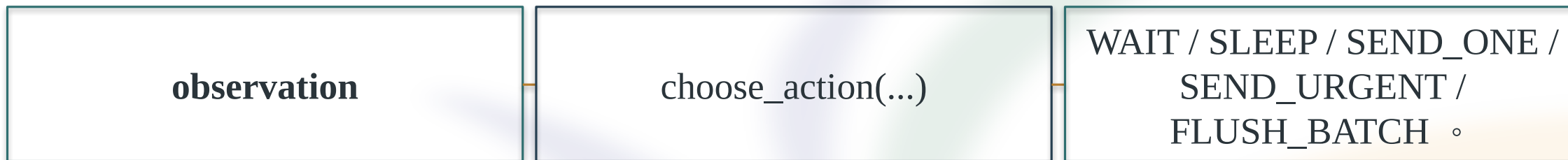
state / receipt



policy API 的輸入是現在能觀察到的事

EVIDENCE PLACEHOLDER

observation card → legal action card ◦



Python survival : 常數是受控旋鈕

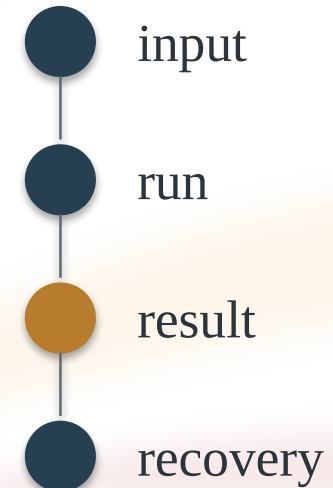
EVIDENCE PLACEHOLDER

copyable operation

PACE_GAP_STEPS 、 REST_DURING_GAP 、 BATCH_SIZE 、 URGENT_MARGIN_S

bounded values only ◦

state / receipt



Python survival : 條件只處理當前 observation

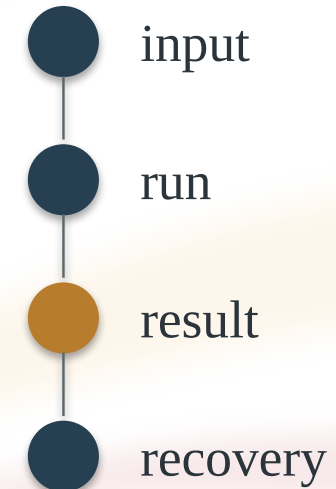
EVIDENCE PLACEHOLDER

copyable operation

現在看到的條件 → 一個 legal action

不要讀 future outcome 。

state / receipt



Python survival : 縮排與 return

EVIDENCE PLACEHOLDER

三行縮排示意與 action token 。

```
choose_action(observ  
ation)
```

```
    return WAIT
```

```
    return SLEEP
```

```
    return SEND_ONE 。
```

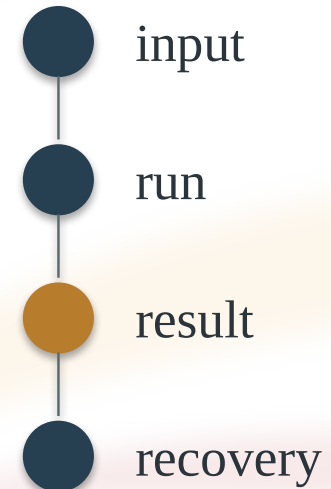
endpoint 狀態：名稱要和機制對齊

VERIFIED

copyable operation

SLEEP 、 WAIT 、 WAKE 、 PROCESS 、 TX 、 RX 。

state / receipt



WAIT 是 awake idle ， SLEEP 會帶來 wake 成本

VERIFIED

WAIT 與 SLEEP 的雙泳道時間線。

WAIT = 保持清醒

two-lane timing

**SLEEP = 低功耗 + wake
latency/energy 。**

VERIFIED

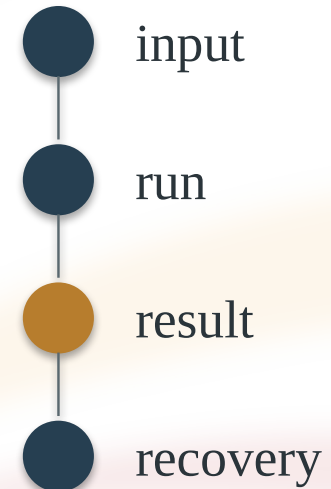
封包生命週期：送出不是交付

VERIFIED

copyable operation

attempt 、 retry 、 delivered 、 expired 、 queue age 。

state / receipt



service boundary 先於 energy 比較

VERIFIED; DONOR-
REWRITE : BeamShift donor
slides 19–21 的 service reaction
與 $W \rightarrow J$ boundary , 改寫成
current endpoint ledger 。

service gate 包住 endpoint energy ledger 。

observation	action	mechanism	evidence
先確認 job / window / delivery	再比較 J 。		

W 是速率，J 是累積結果

VERIFIED; DONOR-
REWRITE : BeamShift donor
slides 14, 16/17 的
throughput、power、energy
，去重後拆成 W 與 J。

P(t) 曲線下的面積。

observation	action	mechanism	evidence
power = rate	energy = area under P(t)	W、J。	

公式： endpoint energy 的範圍

VERIFIED

$$E_{endpoint} = \sum_{s \in S} P_s t_s$$

LaTeX source

$$\eta_E = \frac{D_{delivered}}{E_{endpoint}}$$

公式：服務效率的分子與分母

VERIFIED

$$E_{\text{endpoint}} = \sum_{s \in S} P_s t_s$$
$$\eta_E = \frac{D_{\text{delivered}}}{E_{\text{endpoint}}}$$

scope → numerator / denominator → units → evidence

LaTeX source : $\eta_E = \frac{D_{\text{delivered}}}{E_{\text{endpoint}}}$; delivered bits / J 。

來源、模型、課程假設、結果

VERIFIED

四層 provenance chain ◦

**source → model → course assumption →
result**

scenario_id / seed / policy hash ◦

一個 scenario_id 綁住整條鏈

VERIFIED

runner → result → replay → workbook 的同色 identity token 。

c120-ntpu-energy-decision-01

scenario / anchor / policy / replay 。

公平 baseline : 固定 job 、 window 、 boundary

VERIFIED; DONOR-
REWRITE : BeamShift donor
slides 36–39 的 fair A/B , 改
寫成固定 job/window/boundary
的 bounded policy loop 。

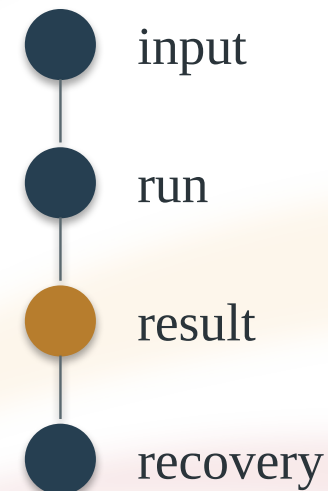
copyable operation

FIXED:scenario / seed / job

CONTROLLED:policy block

OBSERVED:queue / service / state / J 。

state / receipt



LEO window : 合法與不合法的 action

VERIFIED; DONOR-
REWRITE : BeamShift donor
slides 2, 4–6 的 window
opportunity , 改寫成 LEO-only
changing service example 。

contact window 上的 legal / illegal action marks 。

window open

window closed

WAIT / SEND

same anchor 。

Lab A：同一份工作，不同 pace

VERIFIED

同一批 packet 在兩種送出節奏上的雙泳道。

same job

lab hero

pace / rest

VERIFIED

service first ◦

recovery / transfer

先讀 PACE_GAP_STEPS

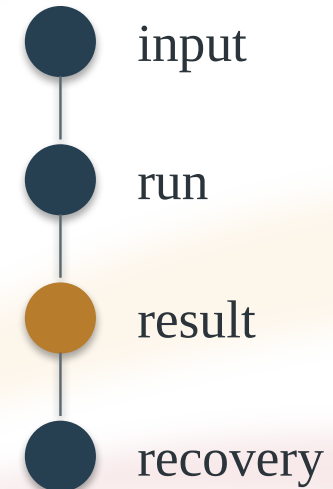
EVIDENCE PLACEHOLDER

copyable operation

gap 變大 → send opportunity 改變

line number:PLACEHOLDER 。

state / receipt



再讀 REST_DURING_GAP

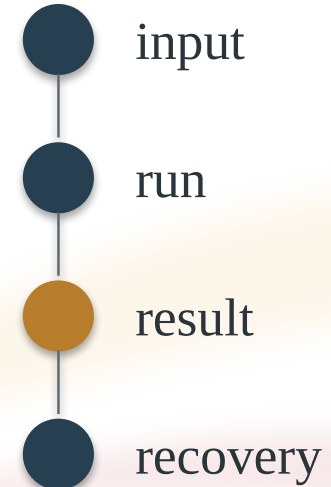
VERIFIED

copyable operation

REST_DURING_GAP = WAIT or SLEEP

wake cost must be observed ◦

state / receipt



Lab A 的 prediction lock

VERIFIED

四格 prediction card 。

queue age 、 service 、 state time 、 endpoint
J

before run 。

A-01 : 跑 untouched baseline

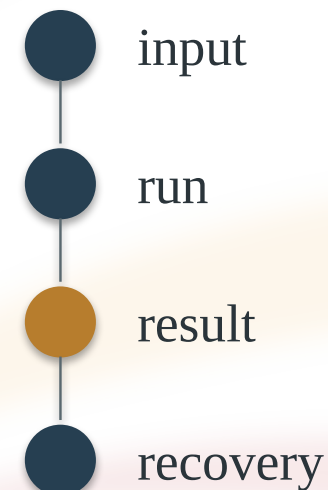
IMPLEMENTED / NOT
VERIFIED

copyable operation

bash course.sh run --lab A --case baseline

Windows:course.cmd run --lab A --case baseline

state / receipt



EXPECT | stdout JSON + result_path

STOP | IMPLEMENTED / NOT VERIFIED ; 保留錯誤

RECOVER | 若 A result 缺失或沒有 consequential diff , 保留 stdout 與錯誤 , rest...

A-02 : 只改 A marked block 並 freeze

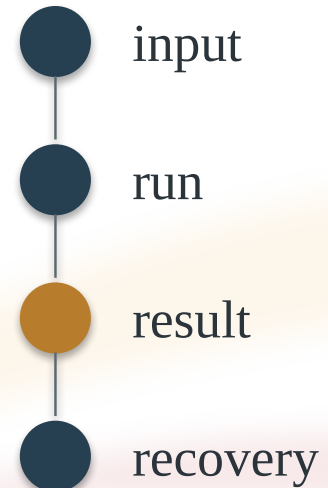
IMPLEMENTED / NOT
VERIFIED

copyable operation

```
bash course.sh run --lab A --case candidate --freeze
```

```
Windows:course.cmd run --lab A --case candidate --freeze
```

state / receipt



EXPECT | stdout JSON + result_path

STOP | IMPLEMENTED / NOT VERIFIED ; 保留錯誤

RECOVER | 若 A result 缺失或沒有 consequential diff , 保留 stdout 與錯誤, rest...

A-03 : 匯入 candidate , 先看同一個 boundary

EVIDENCE PLACEHOLDER

baseline/candidate 雙欄 evidence ledger 。

same scenario

queue / service / state / J
endpoint layer 。

same job

PLACEHOLDER

A-04：讀懂 state ledger

EVIDENCE PLACEHOLDER

WAIT/SLEEP/WAKE/PROCESS/TX/RX 的橫向 ledger 。

observation	action	mechanism	evidence
state duration	wake latency	TX/RX attempts	energy scope 。

A-05：把 packet 結果接回 service

VERIFIED

packet ledger → service gate 的箭頭。

observation	action	mechanism	evidence
attempt ≠ delivered	retry / expired	service verdict。	

A-06 : 檢查 A freeze lineage

EVIDENCE PLACEHOLDER

baseline → candidate → lab-a-frozen receipt 鏈。



EXPECT | policy SHA + freeze receipt

STOP | EVIDENCE PLACEHOLDER ; 不得補填

RECOVER | 若 A result 缺失或沒有 consequential diff , 保留 stdout 與錯誤 , rest...

A-07 : 用 hidden case 檢驗，不再 retune

IMPLEMENTED / NOT
VERIFIED

同一 policy 跨兩個不同窗口的對照圖。

**bash course.sh run --lab A --
case hidden**

Windows:course.cmd run --lab
A --case hidden

policy SHA unchanged 。

EXPECT | stdout JSON + result_path

STOP | IMPLEMENTED / NOT VERIFIED ; 保留錯誤

RECOVER | 若 A result 缺失或沒有 consequential diff，保留 stdout 與錯誤，rest...

Lab A debrief：一條因果句就夠

EVIDENCE PLACEHOLDER

A edit → mechanism → evidence → verdict 四段句卡。

observation	action	mechanism	evidence
我改了…	因此…	result 顯示…	在 hidden…。

Lab B：現在送，還是等到更穩定？

VERIFIED

一條 quality/contact trace 與 send-ready band。

enter

quality trace hero

exit

VERIFIED

hold

recovery / transfer

ENTER_QUALITY : 何時進入 send-ready

EVIDENCE PLACEHOLDER

copyable operation

quality $\geq$ ENTER_QUALITY

enter event

line number:PLACEHOLDER ◦

state / receipt

● input

● run

● result

● recovery

EXIT_QUALITY : 為什麼不能只用一條線

VERIFIED

copyable operation

ENTER_QUALITY $\neq$ EXIT_QUALITY

avoid ping-pong ◦

state / receipt

● input

● run

● result

● recovery

STABLE_STEPS : 拒絕短暫尖峰

EVIDENCE PLACEHOLDER

逐格累積的 stability counter 。

**stable_count <
STABLE_STEPS**

hold

stable_count =
STABLE_STEPS → switch 。

Lab B 的 Trace A prediction

VERIFIED

copyable operation

預測 entry

預測 hold

預測 service / endpoint J 。

state / receipt

● input

● run

● result

● recovery

B-01 : Trace A baseline

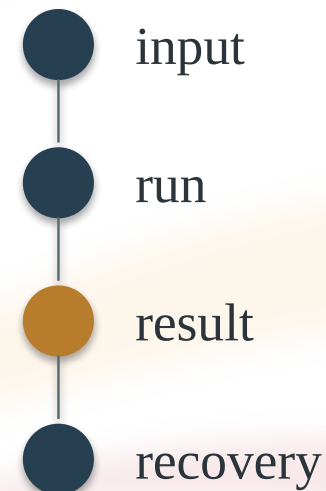
IMPLEMENTED / NOT
VERIFIED

copyable operation

bash course.sh run --lab B --case trace-a-baseline

Windows:course.cmd run --lab B --case trace-a-baseline

state / receipt



EXPECT | stdout JSON + result_path

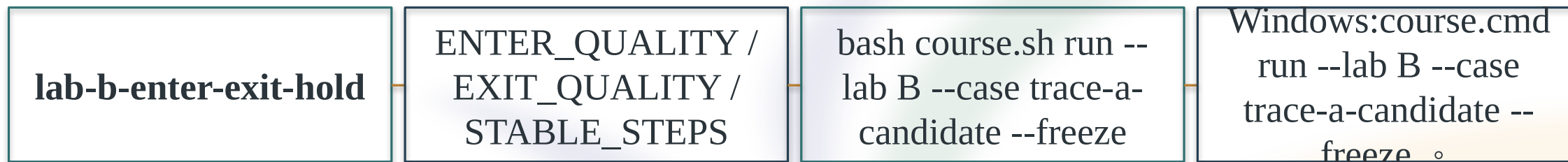
STOP | IMPLEMENTED / NOT VERIFIED ; 保留錯誤

RECOVER | 若 B receipt 、 predecessor 或 policy SHA 不符，停止 withheld ， re...

B-02 : 改 B marked block 並 freeze

IMPLEMENTED / NOT
VERIFIED

三個 B 常數中的單一 diff 。



EXPECT | stdout JSON + result_path

STOP | IMPLEMENTED / NOT VERIFIED ; 保留錯誤

RECOVER | 若 B receipt 、 predecessor 或 policy SHA 不符，停止 withheld ， re...

B-03：讀 transition 與 service，不只讀品質

EVIDENCE PLACEHOLDER

mode transition 與 packet/service 的同步表。

observation	action	mechanism	evidence
quality trace	mode transition	attempt / retry	service

B-04：確認 freeze receipt 才能進 Trace B

EVIDENCE PLACEHOLDER

B frozen receipt 的鎖與 predecessor 箭頭。

observation	action	mechanism	evidence
lab-b-frozen	policy SHA	Trace A	predecessor = A frozen。

EXPECT | policy SHA + freeze receipt
STOP | EVIDENCE PLACEHOLDER；不得補填
RECOVER | 若 B receipt、predecessor 或 policy SHA 不符，停止 withheld，re...

B-05 : Trace B 不 retune

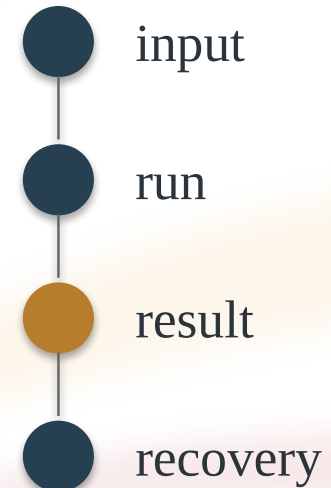
IMPLEMENTED / NOT
VERIFIED

copyable operation

```
bash course.sh run --lab B --case trace-b
```

```
Windows:course.cmd run --lab B --case trace-b
```

state / receipt



EXPECT | stdout JSON + result_path

STOP | IMPLEMENTED / NOT VERIFIED ; 保留錯誤

RECOVER | 若 B receipt 、 predecessor 或 policy SHA 不符，停止 withheld ， re...

B-06：辨識 too-slow 與 ping-pong

VERIFIED; DONOR-REWRITE：BeamShift donor slide 12 的 too-slow / ping-pong counterexample，改寫成 enter/exit/hold evidence。

兩條 transition traces。

too-slow

service loss
保留 counterexample。

ping-pong

VERIFIED; DONOR-REWRITE：
BeamShift donor slide 12 的 too-slow /
ping-pong counterexample，改寫成
enter/exit/hold evidence。

Lab B debrief : hysteresis 是條件，不是保證

EVIDENCE PLACEHOLDER

policy condition → transition → service/energy → verdict 。

成立

verdict card

被推翻

PLACEHOLDER

待查。

recovery / transfer

Recovery checkpoint：保存 A/B lineage

EVIDENCE PLACEHOLDER

A frozen、B frozen 與 workbook checkpoint 三層堆疊。

observation	action	mechanism	evidence
artifacts/receipts/	artifacts/checkpoints/	workbook prediction + result。	

Lab C：佇列、批次、休眠或 urgent send

VERIFIED

queue cards 接到長窗口與短窗口。

batch

queue hero

urgent

VERIFIED

deadline

recovery / transfer

queue 、 deadline 、 freshness 不是同一欄

VERIFIED

queue age 、 deadline 、 freshness 三張訊息卡。

queue age

freshness
delivered / expired ◦

deadline

VERIFIED

BATCH_SIZE : 少 activation 還是多等待

EVIDENCE PLACEHOLDER

copyable operation

BATCH_SIZE

flush

activation count

state / receipt

● input

● run

● result

● recovery

URGENT_MARGIN_S：何時跳過一般節奏

EVIDENCE PLACEHOLDER

copyable operation

$\text{deadline} - \text{now} \leq \text{URGENT_MARGIN_S}$

SEND_URGENT

不要偷看結果。

state / receipt

● input

● run

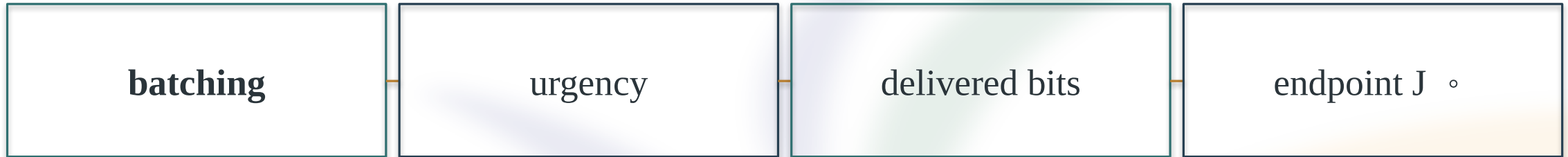
● result

● recovery

Lab C prediction : 先寫 trade-off

VERIFIED

activation $\leftrightarrow$ delay $\leftrightarrow$ service $\leftrightarrow$ J 的四向平衡圖。



C-01 : 跑 baseline

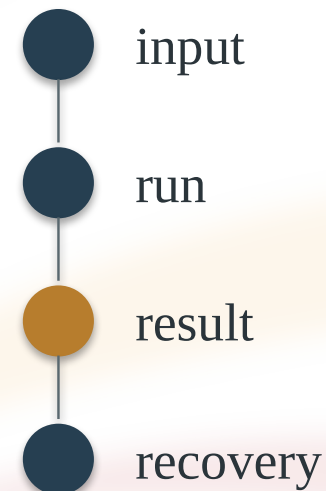
IMPLEMENTED / NOT
VERIFIED

copyable operation

```
bash course.sh run --lab C --case baseline
```

```
Windows:course.cmd run --lab C --case baseline 。
```

state / receipt



EXPECT | stdout JSON + result_path

STOP | IMPLEMENTED / NOT VERIFIED ; 保留錯誤

RECOVER | 若 C lineage 、 packet ledger 或 service evidence 不完整，保留目前 ...

C-02 : 只做一次 candidate edit

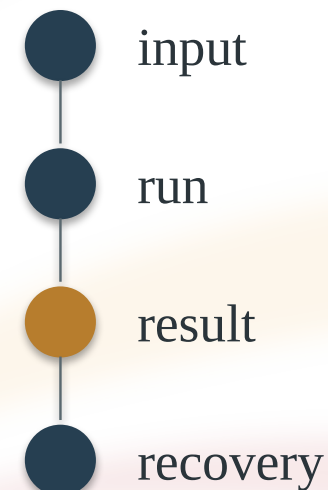
IMPLEMENTED / NOT
VERIFIED

copyable operation

bash course.sh run --lab C --case candidate

Windows:course.cmd run --lab C --case candidate ◦

state / receipt



EXPECT | stdout JSON + result_path

STOP | IMPLEMENTED / NOT VERIFIED ; 保留錯誤

RECOVER | 若 C lineage 、 packet ledger 或 service evidence 不完整，保留目前 ...

C-03：讀 queue 與 packet ledger

EVIDENCE PLACEHOLDER

queue cards 與 delivered/expired packet ledger。

observation	action	mechanism	evidence
queue age	attempt	retry	delivered

C-04 : 讀 state 與 endpoint J

EVIDENCE PLACEHOLDER

copyable operation

state duration

$P_s \times t_s$

E_{endpoint}

state / receipt

● input

● run

● result

● recovery

C-05：根據 evidence 做唯一一次 revision

VERIFIED

candidate evidence → 一個 revision arrow。

observation	action	mechanism	evidence
one reason	one marked-block change	no second revision。	

C-06 : revision freeze

IMPLEMENTED / NOT
 VERIFIED

C candidate checkpoint → C frozen receipt ◦

observation	action	mechanism	evidence
bash course.sh run --lab C --case revision --freeze	Windows:course.cmd run --lab C --case revision --freeze	lab-c-frozen	policy SHA ◦

EXPECT | stdout JSON + result_path
 STOP | IMPLEMENTED / NOT VERIFIED ；保留錯誤
 RECOVER | 若 C lineage 、 packet ledger 或 service evidence 不完整，保留目前 ...

C-07 : surprise case 保持 frozen policy

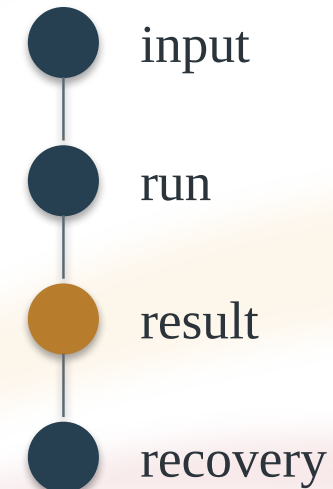
IMPLEMENTED / NOT
VERIFIED

copyable operation

```
bash course.sh run --lab C --case surprise
```

```
Windows:course.cmd run --lab C --case surprise
```

state / receipt



EXPECT | stdout JSON + result_path

STOP | IMPLEMENTED / NOT VERIFIED ; 保留錯誤

RECOVER | 若 C lineage 、 packet ledger 或 service evidence 不完整，保留目前 ...

Lab C debrief：用 packet、service、J 一起判讀

EVIDENCE PLACEHOLDER

candidate / revision / surprise 三欄 ledger。

observation	action	mechanism	evidence
packet outcome	service	endpoint J	counterexample。

先保存 result 與配對 replay

IMPLEMENTED / NOT
VERIFIED

result/replay 雙檔案與同一 run_id 的連結。

observation	action	mechanism	evidence
artifacts/ <run_id>/ result.json	artifacts/<run_id>/ endpoint-replay.json	stdout result_path 。	

在 Leo 先記 setup mode

EVIDENCE PLACEHOLDER

read the mechanism

READY receipt → 記錄
same-scenario fallback → 記錄
fallback。

SERVER PREVIEW / FIXTURE HOST ONLY

strict importer 檢查 schema、scenario identity/hash、policy、seed、units 與 provenance，通過後才建立 endpoint replay。 / The strict importer checks schema, scenario identity/hash, policy, seed, units, and provenance before creating an endpoint replay.

預期 / Expect
成功時一次接受並更新 import ledger、replay 與 workbook evidence；失敗時 session 不變。 / Success accepts the file once and updates the import ledger, replay, and workbook evidence; failure leaves the session unchanged.

如何解讀 / recover / Interpret / recover
只接受 student-run 或有標籤的 same-scenario fallback；mismatch 就修正檔案或回到 matching artifact，不要放寬 gate。 / Accept only a student-run or labelled same-scenario fallback; for a mismatch, fix the artifact or return to the matching one instead of weakening the gate.

匯入 result.json / Import result.json 只選擇檔案；元件不在瀏覽器解析。 / Choose a file; this component does not parse it in the browser. No file chosen

匯入狀態 / Import status: 尚未匯入 endpoint result。 / No endpoint result imported yet.

ENDPOINT REPLAY / ENDPOINT REPLAY
匯入通過後才會顯示 queue、packet、radio state 與 endpoint energy。 / Queue, packet, radio-state, and endpoint-energy evidence appears after a validated import.

RUN LEDGER 與比較 / RUN LEDGER AND COMPARISON
尚未有 baseline、candidate、revision 或 withheld artifact。 / No baseline, candidate, revision, or withheld artifact has arrived yet.

EXPECT | READY receipt

STOP | EVIDENCE PLACEHOLDER；不得補填

RECOVER | 若 setup mode 與 artifact source 不一致，停止 import，先改回相符的 RE...

匯入 exact result ，不上傳 policy source

EVIDENCE PLACEHOLDER

read the mechanism

匯入 result.json
fresh:artifacts/<run_id>/...
fallback:fallback_artifacts/... 。

SERVER PREVIEW / FIXTURE HOST ONLY

/ Accept only a student-run or labelled same-scenario fallback; for a mismatch, fix the artifact or return to the matching one instead of weakening the gate.

匯入 result.json / Import result.json 只選擇檔案；元件不在瀏覽器解析。 / Choose a file; this component does not parse it in the browser. result.json

匯入狀態 / Import status: 已完整匯入 A/baseline；session 只在所有 gate 通過後更新。 ·
run_id=sha256:5d2e38ebaabaa0433bfb50a723472fcc2eed5adbe7bfa423c00f15970fafd8

same-scenario-fallback

只讀 endpoint layer · 不改變 C-120 system replay / Read-only endpoint layer · does not rewrite C-120 system replay
QUEUE · PACKET 與 RADIO-STATE REPLAY / QUEUE, PACKET, AND RADIO-STATE REPLAY
目前 frame 由 controller 提供；瀏覽器只呈現 artifact 值，不重新套用科學公式。 / The controller supplies the current frame; the browser displays artifact values without reapplying scientific formulas.

上一格 / Previous frame 選擇 endpoint replay frame / Select endpoint replay frame 第 1 格 · 0s / Frame 1 · 0s

目前 frame / Current frame 1/56 · 0s · 無接觸 / no contact · quality 0

佇列 Queue / Queue 佇列 · normal-1	封包事件 Packet events / Packet events 封包事件	Endpoint radio state / Endpoint radio state
-----------------------------------	---	---

EXPECT | consequential diff across state / packet / service / J

STOP | EVIDENCE PLACEHOLDER ；不得補填

RECOVER | 若匯入被拒絕，保留 importer error ，核對 result/replay pair 與 case ，...

importer 逐層檢查 provenance

VERIFIED; DONOR-
REWRITE : BeamShift donor
slides 70-75 的 evidence
qualification , 改寫為 current
schema 、 identity 、 unit 、 poli
cy 與 provenance gate 。

六階 validation ladder , 最後才落到 replay 。

schema → identity → units → policy / seed → provenance → replay 。

EXPECT | consequential diff across state / packet / service / J

STOP | identity / unit / service 不一致就停止

RECOVER | 任一 gate 失敗就回到命名的 artifact 或 same-scenario fallback

replay 把 result 變成可讀時間線

EVIDENCE PLACEHOLDER

read the mechanism

queue
packet
radio state
contact / handover ; endpoint
energy 。

SERVER PREVIEW / FIXTURE HOST ONLY

/ Accept only a student-run or labelled same-scenario fallback; for a mismatch, fix the artifact or return to the matching one instead of weakening the gate.

匯入 result.json / Import result.json 只選擇檔案；元件不在瀏覽器解析。 / Choose a file; this component does not parse it in the browser. result.json

匯入狀態 / Import status: 已完整匯入 A/baseline；session 只在所有 gate 通過後更新。 ·
run_id=sha256:5d2e38ebaabaaa0433bfb50a723472fcc2eed5adbe7bfa423c00f15970fafd8

same-scenario-fallback

只讀 endpoint layer · 不改變 C-120 system replay / Read-only endpoint layer · does not rewrite C-120 system replay
QUEUE · PACKET 與 RADIO-STATE REPLAY / QUEUE, PACKET, AND RADIO-STATE REPLAY
目前 frame 由 controller 提供；瀏覽器只呈現 artifact 值，不重新套用科學公式。 / The controller supplies the current frame; the browser displays artifact values without reapplying scientific formulas.

上一格 / Previous frame 選擇 endpoint replay frame / Select endpoint replay frame 第 1 格 · 0s / Frame 1 · 0s ▾

目前 frame / Current frame 1/56 · 0s · 無接觸 / no contact · quality 0

佇列 Queue / Queue 佇列 • normal-1	封包事件 Packet events / Packet events 封包事件	Endpoint radio state / Endpoint radio state
-----------------------------------	---	---

server preview 可以證明什麼

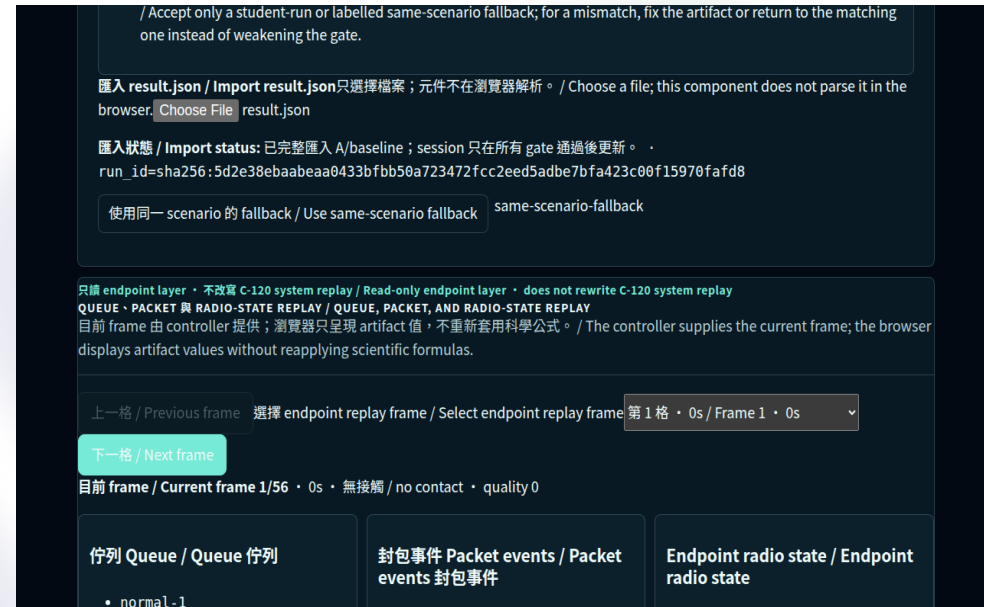
read the mechanism

Server preview 已驗證
Windows native:PLACEHOLDER
endpoint KPI:PLACEHOLDER。

VERIFIED (server preview /
fixture host boundary only) ;
DONOR-REWRITE :

BeamShift donor slides 96–97 的
claim boundary , 改寫成
server-preview 與
Windows / WISE 的明確分層。

SERVER PREVIEW / FIXTURE HOST ONLY



匯出可重開的 Energy Decision Workbook

EVIDENCE PLACEHOLDER

read the mechanism

匯出可重開的學習單
predictions
run ledger
replay IDs ; source mode 。

SERVER PREVIEW / FIXTURE HOST ONLY



關閉再 reopen ，驗證 provenance 仍在

EVIDENCE PLACEHOLDER

read the mechanism

export → close → reopen
same scenario_id
same run / replay records 。

SERVER PREVIEW / FIXTURE HOST ONLY



EXPECT | workbook reopens with provenance

STOP | EVIDENCE PLACEHOLDER ；不得補填

RECOVER | 若 identity 不符或 evidence 缺失，保留 INCOMPLETE ，回到同一 scenario...

mismatch 必須 fail closed

VERIFIED (server preview
mismatch recovery)

read the mechanism

identity / unit / schema mismatch
session unchanged
REJECTED 。

SERVER PREVIEW / FIXTURE HOST ONLY

creating an endpoint replay.
預期 / Expect
成功時一次接受並更新 import ledger、replay 與 workbook evidence；失敗時 session 不變。 / Success accepts the file once and updates the import ledger, replay, and workbook evidence; failure leaves the session unchanged.
如何解讀 / recover / Interpret / recover
只接受 student-run 或有標籤的 same-scenario fallback；mismatch 就修正檔案或回到 matching artifact，不要放寬 gate。
 / Accept only a student-run or labelled same-scenario fallback; for a mismatch, fix the artifact or return to the matching one instead of weakening the gate.

匯入 result.json / Import result.json 只選擇檔案；元件不在瀏覽器解析。 / Choose a file; this component does not parse it in the browser. c120-mismatch-result.json

匯入狀態 / Import status: 拒絕匯入；目前 session 未變更。 schema:CONSTANT_MISMATCH \$.scenario_id — expected "c120-ntpu-energy-decision-01"

只讀 endpoint layer • 不改變 C-120 system replay / Read-only endpoint layer • does not rewrite C-120 system replay
QUEUE、PACKET 與 RADIO-STATE REPLAY / QUEUE, PACKET, AND RADIO-STATE REPLAY
目前 frame 由 controller 提供；瀏覽器只呈現 artifact 值，不重新套用科學公式。 / The controller supplies the current frame; the browser displays artifact values without reapplying scientific formulas.

上一格 / Previous frame 選擇 endpoint replay frame / Select endpoint replay frame 第 1 格 • 0s / Frame 1 • 0s

EXPECT | reject; no partial update

STOP | identity / unit / service 不一致就停止

RECOVER | 保存錯誤訊息，核對 manifest、scenario、case 與配對 replay

fallback 是同一場景的可追溯復原

EVIDENCE PLACEHOLDER

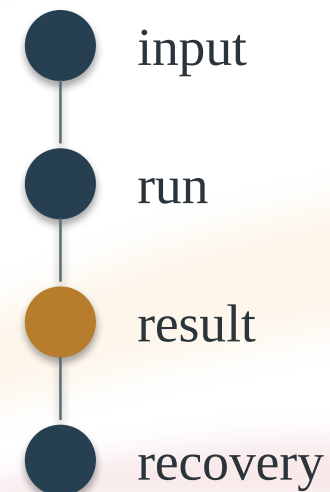
copyable operation

artifact_source: same-scenario-fallback

manifest.json

matching result + replay ◦

state / receipt



policy guard 失敗：只恢復 marked block

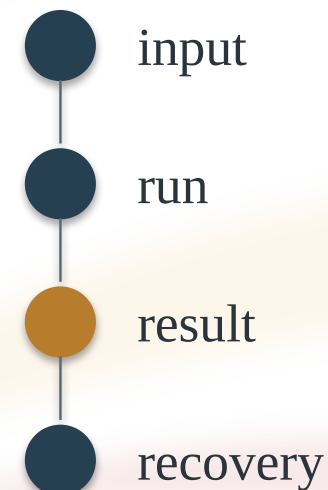
IMPLEMENTED / NOT
VERIFIED

copyable operation

status → restore --checkpoint → policy guard

artifacts untouched ◦

state / receipt



EXPECT | consequential diff across state / packet / service / J

STOP | IMPLEMENTED / NOT VERIFIED ; 保留錯誤

RECOVER | checkpoint 不存在或 identity 不符，就回到 release-default 或 matc...

一條完整的中斷復原路徑

VERIFIED

從 runner error 到 workbook reopen 的單一路徑。

保留錯誤 → 核對 identity → restore / fallback → import → reopen 。

EXPECT | workbook reopens with provenance

STOP | identity / unit / service 不一致就停止

RECOVER | 若 recovery provenance 無法說清楚，將本次 session 標成 INCOMPLETE ，...

evidence clinic : 決策時可取得什麼

VERIFIED

copyable operation

decision-time input

post-action outcome

不可互換。

state / receipt

● input

● run

● result

● recovery

leakage 會讓假說看起來很強

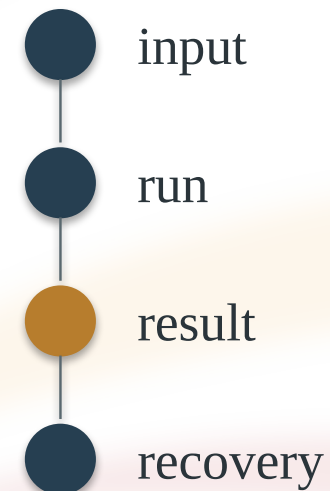
VERIFIED

copyable operation

合法：現在的 quality / queue / deadline

禁止：future delivery / future energy。

state / receipt



一筆 evidence record 要能自我說明

VERIFIED; DONOR-
REWRITE : BeamShift donor
slides 70-75、96-97、113 的
evidence record 與
qualification，改寫成 current
artifact / claim record。

一筆 record 卡片串起 source、policy、result、replay 與 verdict。

observation	action	mechanism	evidence
source	identity	mechanism	evidence

measured 、 derived 、 assumed 、 simulated

VERIFIED; DONOR-
REWRITE : BeamShift donor
slide 113 的四類 claim , 改寫
為 current course boundary 。

copyable operation

measured

derived

assumed

state / receipt

● input

● run

● result

● recovery

endpoint layer 與 system layer 的邊界

VERIFIED; DONOR-
REWRITE : BeamShift donor
evidence-boundary concepts , 改
寫為 endpoint replay / C-120
system replay 分層。

兩個有明確邊界的 energy ledgers 。

observation	action	mechanism	evidence
endpoint J 不等 於 system consumed J	共享 scenario ≠ 共 享語義。		

把同一機制轉到智慧農場

VERIFIED

copyable operation

soil sensor

上行機會

batch / sleep / urgent

state / receipt

● input

● run

● result

● recovery

HVAC 、 edge 、 物流：機會變動仍是同一個問題

VERIFIED

三個 domain card 共同指向一條 causal spine 。

HVAC

logistics
opportunity → policy → service / energy 。

edge inference

VERIFIED

寫一個可被推翻的競賽假說

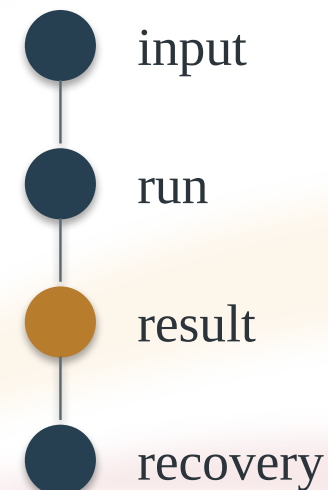
VERIFIED

copyable operation

在固定 _____ 下，若改變 _____，則 _____ 會改變，
因為 _____

若 _____，假說被推翻。

state / receipt



Appendix : TLE 只負責提供時間與位置來源

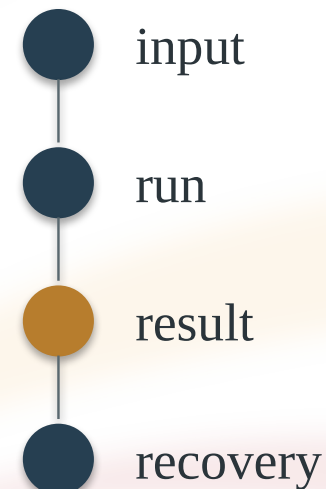
VERIFIED; DONOR-
REWRITE : BeamShift donor
slides 7-10 、 98-110 , 僅取
TLE/SGP4/coordinate
provenance vocabulary 。

copyable operation

pinned source → model → contact context

LEO = changing window example 。

state / receipt



Appendix : SGP4 與 contact window 的最小讀法

VERIFIED

position sample 點穿過 contact boundary 的圖。

position over time

contact open / closed

not live telemetry 。

Appendix : quality 、 dB 與 action context

VERIFIED; DONOR-
REWRITE : BeamShift donor
slides 13 、 15 、 22-35 、 111-
114 , 縮成非通訊背景可讀的
quality vocabulary 。

dB observation 透過 threshold 進入 policy 的最小流程。

quality observation $\neq$ service

dB / linear

不要求 link-budget 推導。

Appendix : W 、 J 、 bit/s 、 delivered bits 、 bit/J

VERIFIED; DONOR-
REWRITE : BeamShift donor
slides 14 、 16/17 ， 去重後重建
單位階梯。

五個單位 token 從 state ledger 接到 service gate 。

observation	action	mechanism	evidence
W = power rate	J = accumulated energy	bit/s = rate	delivered bits

Appendix : source → model → assumption → result

VERIFIED

provenance spine 與四個可追溯節點。



Appendix : student_policy.py 的 allowed API

EVIDENCE PLACEHOLDER;
DONOR-REWRITE :

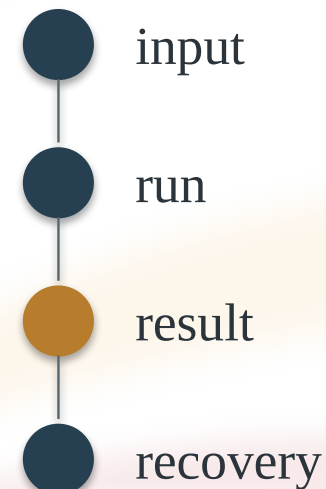
BeamShift 舊操作 / session 語
義退役，僅保留 current
bounded action contract 。

copyable operation

allowed observations → WAIT / SLEEP / SEND_ONE /
SEND_URGENT / FLUSH_BATCH

API version:PLACEHOLDER 。

state / receipt



Appendix : scenario 、 result 、 receipt 、 replay 的契約流

EVIDENCE PLACEHOLDER

五份 schema 連成一條 contract flow 。

scenario.schema → result.schema → freeze-receipt.schema → endpoint-replay.schema →
workbook 。

EXPECT | consequential diff across state / packet / service / J

STOP | EVIDENCE PLACEHOLDER ；不得補填

RECOVER | schema validation 失敗時，保留 error 與 input ，不手改欄位，改選 matchi...

Appendix : fail-closed validation ladder

VERIFIED

每一關通過才向下的閘門。

shape → identity → units → policy →
provenance

fail = no partial update ◦

Appendix : 課程 wrapper 與 upstream license boundary

VERIFIED

copyable operation

course wrapper

pinned upstream reference

JSON seam

state / receipt

● input

● run

● result

● recovery

Appendix : field glossary 與 claim ceiling

VERIFIED

glossary tokens 與 claim shield 。

queue

glossary card

packet

VERIFIED

service

recovery / transfer

快速分支：同一 policy，新的 held-out 條件

VERIFIED

frozen policy 直接連到新的 counterexample card。

freeze policy

predict → run → compare
no retune。

new opportunity / trace

VERIFIED

EXPECT | consequential diff across state / packet / service / J

STOP | identity / unit / service 不一致就停止

RECOVER | 若沒有 frozen receipt 或新的 condition 未標 identity，就回到主路徑，使用…

退出：留下可重開、可反駁的 evidence chain

EVIDENCE PLACEHOLDER

workbook 、 hypothesis 、 falsifier 三張卡回到同一 scenario identity 。

export

exit gate

close

PLACEHOLDER

reopen

recovery / transfer

EXPECT | workbook reopens with provenance

STOP | EVIDENCE PLACEHOLDER ；不得補填

RECOVER | 若無法完成 export/reopen ，保留已產生的 artifact 與錯誤，使用 backup 或 sa...